Mandatory deliverable

LAB | Resize and Attach EBS Volumes in AWS

Lab

Skills you will learn

Cloud Computing: Storage & Databases

Introduction

In this hands-on lab, you will create an EC2 instance, attach an additional EBS volume, and resize both the root EBS volume and the attached volume. You will also extend the filesystem to reflect the updated sizes.

Objectives

By the end of this lab, you will be able to:

1. Launch an EC2 instance and configure EBS storage.
2. Attach an additional EBS volume to the EC2 instance.
3. Modify and resize the root volume.
4. Extend the filesystem to use the newly allocated space.
5. Verify volume changes and ensure they reflect correctly.
6. Clean up AWS resources to optimize costs.

Lab Instructions

1. Create an EC2 Instance

- **Launch a new EC2 instance** named `resize-ec2`:
  - **AMI:** Amazon Linux 2 (default AMI).
  - **Instance Type:** `t2.micro`.
  - **Key Pair:** Proceed without a key pair (or use an existing one).
  - **Storage Configuration:**
    - Set the **root volume** to `10 GiB`.

- **Add an additional EBS volume** of `10 GiB`.

- **Launch the instance.**

2. Verify Attached Volumes

- **Connect to the EC2 instance** using **EC2 Instance Connect**.
- **List available block devices** to confirm the attached volumes:

```
lsblk
```

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- **Expected output:** You should see `xvda` (root volume) and `xvdb` (additional volume).

3. Resize the Root Volume

- **Modify the root volume size** from `10 GiB` to `20 GiB`:
  1. Navigate to the **EC2 console**.
  2. Select **Volumes** in the left panel.
  3. Choose the root volume (`/dev/xvda`) and click **Modify Volume**.
  4. Set the **new size** to `20 GiB` and confirm.
  5. Wait for the modification to complete (check volume state in AWS console).

4. Extend the Root Partition and Filesystem

- **Reconnect to the instance** via EC2 Instance Connect.
- **Verify the root partition size:**

```
lsblk
```

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- **Resize the root partition:**

```
sudo growpart /dev/xvda 1
```

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- **Verify partition resize:**

```
lsblk
```

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- **Check filesystem size before extension:**

```
df -hT
```

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- **Extend the filesystem:**

```
sudo xfs_growfs -d /
```

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(If using ext4, run: `sudo resize2fs /dev/xvda1` instead.)

- **Verify the new filesystem size:**

```
df -hT
```

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5. Cleanup Resources

- **Terminate the EC2 instance**.
- **Delete the attached EBS volumes** to avoid unnecessary charges.

Completion Criteria

- EC2 instance `resize-ec2` created with two EBS volumes.
- Root EBS volume successfully resized from `10 GiB` to `20 GiB`.
- Filesystem extended and verified using `df -hT`.
- All resources cleaned up after lab completion.

Submission Guidelines

- Take **screenshots** of each major step.
- Save them into a **document (Google Docs, PDF, etc.)**.
- Upload to **Google Drive** and share the link.
- Ensure **instructor has access** before submission.

🚀 **Great job! You've successfully resized and attached EBS volumes in AWS!** 🎉

LAB | Resize and Attach EBS Volumes in AWS

Lab

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